

Model Structure (25.12_1Reg_Upd_Growth):

This is a panmictic model (do_spatial == 0).

There are 2 sexes, 2 fishing fleets, 3 survey fleets, and 1 regions being modeled.

The model has 66 years, starting in 1960 and ending in 2025.

There are 30 ages, where recruitment occurs at age-2 and the plus group occurs at age-31.

There are 30 length bins, which start at 41cm and end at 99cm.

Growth Estimation:

This model estimates new growth curves for the post-1996 time block (upd_growth == 1, use_new_bio == 2) and uses the 2021 SAFE growth estimates for the pre-1996 time block (use_hist_grwth_blk == 1, do_rec_growth == 0). A single time block of weight is used from the 2021 SAFE (Echave et al., 2021) (use_new_bio == 2).

For fishery composition data aggregation catch-weighting, the historic sex-aggregated Weight parameters are used to convert to catch in numbers (sex_disagg_comp_wt == 0).

Aleutian Islands (AI) Data Use:

This model uses data from the Aleutian Islands (AI) -- no data is being removed (drop_AI == 0).

Longline Survey Index Construction:

There was a longline survey in 2025, so there is a terminal year LLS RPN value (LLS_active == 1).

The extrapolated values for the western AI longline survey stations WERE included in the LLS RPN index (drop_WAI_LLS == 0).

The longline survey used off-year extrapolations for the BS (even years) or AI (odd years) based on the year over year change in the GOA RPNs.

Starting in 2025, carryover values for the BS+AI (odd years) or GOA (even years) were used (drop_LLS_extrap == 0 & do_LLS_carry == 0).

The SPoRC model is being passed the index SEs, no conversions needed (do_cont_SE == 1).

Composition Data Structure and ISS:

Composition proportions by age/length are aggregated across regions after weighting by catch in numbers or RPNs.

Age and length compositions are fit 'joint' by sex, but lengths are dropped for years when ages exist (JPN LLS lengths are kept for all years)

(do_spatial == 0 & sex_age_comps == 5).

The Input Sample Size (ISS) for composition data is fixed across years at 20

(do_spatial == 0 & do_scale_ISS == 0).

Length Composition Data Use:

The longline survey length composition data is weighted/calculated using the pre-calculated length RPN data from AKFIN (this is traditional approach) (LLS_length_comp_source == 0).

Longline survey length data are fit in the model as determined by the sex_age_comps switch BUT terminal in-year length data are NOT fit

(fit_LLS_len == 1 & fit_LLS_len_term == 0).

Fixed gear fishery length data are fit in the model as determined by the sex_age_comps switch (fit_LLF_len == 1).

General Data Fitting Notes:

Fishery whale depredation estimates were fixed/static at the previous year's values (whale_dep_static == 1).

The GOA trawl survey data was NOT fit in this model (use_trwl_srvy == 0).

The fishery CPUE index WAS fit in this model (terminal year 2022) using the STANDARDIZED index values (Cheng et al., 2023; ICES JMS) (use_CPUE == 1 & use_CPUE_stand == 1).

Non-fishery catch (e.g., scientific removals) WAS included in the model as fixed gear catch, where only LLS removals were available for terminal year (add_non_fish_catch == 1 & non_fish_catch_updated == 0).